

25.	A	According to Faraday's law, $E = -\frac{dN\Phi}{dt}$. Magnitude of induced e.m.f. in coil = $= \frac{B\Delta A}{\Delta t} = \frac{5.00(\pi(8.0 \times 10^{-2})^2)}{10.0 \times 10^{-3}} = 10.1 \text{ V}$ Since resistance of coil is $4.00 \text{ } \Omega$, current through coil = $10.1 / 4.00 = 2.51 \text{ A}$ According to Lenz's law, direction of induced e.m.f. is to oppose the change causing it. Thus the direction of induced e.m.f. will produce a current to oppose the decreasing flux into the coil. Hence by right hand grip rule, the current will be flowing clockwise in order to produce flux going into the page.
26.	A	Minimum time taken for the current to reach $100 \times 10^{-3} = 0.1 \text{ A}$